Predictive Analysis on Hospital Readmission of Diabetic Patients

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Abstract—Hospital readmission among diabetic patients is a major challenge for healthcare systems worldwide, contributing to increased medical costs, resource utilization, and poor patient outcomes. Predicting readmission risk using large clinical datasets can support early intervention and improve quality of care. This study focuses on predictive analysis of diabetic patient readmission using the Diabetes 130 US Hospitals dataset, which contains over 100,000 hospital encounters collected over a ten-year period. The dataset includes patient demographics, laboratory results, medications, diagnoses, and hospitalization history. Machine learning techniques can be applied to identify patterns and risk factors associated with early readmission. The findings of this research aim to support data-driven decisionmaking, enhance discharge planning, and reduce preventable hospital readmissions. This study demonstrates the potential of predictive analytics and artificial intelligence in improving healthcare efficiency and patient management.

I. Introduction

Diabetes is one of the most common long-term diseases in the world. Many people with diabetes are admitted to hospitals because of complications like infections, heart problems, or sudden changes in blood sugar levels. One major concern in healthcare is hospital readmission, which happens when a patient returns to the hospital soon after being discharged. Readmission within 30 days is often used to measure how effective the treatment and follow-up care were. Predicting which patients are likely to be readmitted is very important. If doctors and hospitals can identify high-risk patients early, they can provide better care, proper follow-up, and guidance after discharge. This can help patients stay healthier and reduce unnecessary hospital visits. Today, hospitals collect large amounts of patient data through electronic health records. Using this data, predictive analysis and machine learning can help find patterns and risk factors that may lead to readmission. This study uses the Diabetes 130-US Hospitals dataset, which contains detailed information about hospital visits of diabetic patients over a ten-year period. The dataset includes information such as patient age, diagnosis, treatments, lab results, and previous hospital visits. By analyzing this data, the study aims to understand what factors increase the chances of hospital readmission among diabetic patients. Many modern healthcare studies, including research published by Elsevier, show that data analysis and artificial intelligence can help doctors make better decisions. Understanding why patients return to the hospital can help improve treatment plans, support better patient care, and reduce healthcare costs.

II. OBJECTIVES

1.To analyse the Diabetes 130-US Hospitals dataset and understand the characteristics of diabetic patients who are readmitted to the hospital.

2.To identify the major factors that contribute to hospital readmission among diabetic patients, such as age, medical conditions, treatments, and previous hospital visits.

3.To apply predictive analysis or machine learning techniques to estimate the likelihood of patient readmission.

4.To evaluate the performance of prediction models in identifying high-risk patients who may need extra care after discharge.

5.To provide insights that can help improve patient care and reduce hospital readmission rates through better monitoring, treatment planning, and follow-up support.

III. LITERATURE REVIEW

[1]Recent studies emphasize that diabetes mellitus significantly increases 30-day hospital readmission rates, contributing to rising healthcare costs. Research using large EHR datasets, such as the Health Facts Database, has enabled the application of machine learning models to predict readmission risks. Key predictors identified include age, number of inpatient admissions, diagnosis count, emergency visits, and length of stay. Comparative analyses show that Random Forest outperforms Naive Bayes and decision tree models, achieving higher AUC values in readmission prediction. Overall, literature supports the use of ML-based predictive models to identify high-risk diabetic patients and improve clinical decision-making and resource allocation.

[2]Recent literature highlights the growing use of machine learning techniques to predict hospital readmission and length of stay (LOS) among diabetic patients. Large-scale datasets from multiple hospitals have enabled the development of robust predictive models using advanced preprocessing and feature selection methods. Studies identify key predictors such as age, number of diagnoses, and prior hospital encounters as

significant factors influencing readmission risk. Validation approaches like cross-validation and hyperparameter tuning improve model reliability and generalizability. Among various algorithms, Random Forest frequently demonstrates superior performance, supporting the role of predictive analytics in enhancing clinical decision-making and healthcare resource management.

[3] Literature indicates that hospital readmission is a significant quality and cost concern, particularly among patients with diabetes. Studies show that diabetic patients have higher readmission rates due to factors such as comorbidities, socioeconomic disparities, and prior hospitalizations. Research highlights that racial/ethnic minorities and individuals with public insurance are at greater risk. Evidence suggests that interventions like inpatient education, improved discharge planning, and coordinated post-discharge care may help reduce readmissions. However, existing studies emphasize the need for more targeted research to evaluate the effectiveness of these interventions specifically for diabetic populations.

IV. PROBLEM STATEMENT:

Hospital readmission among diabetic patients is a major concern for healthcare systems because it increases medical costs, affects hospital resources, and may indicate gaps in patient care after discharge. Many patients with diabetes are readmitted within a short period due to complications, poor disease management, or lack of proper follow-up care. Identifying patients who are at high risk of readmission is difficult because it depends on many factors such as age, medical history, treatment patterns, and existing health conditions. Traditional methods of identifying readmission risk often rely on limited clinical judgment and may not fully use the large amount of data available in hospital records. With the growing use of electronic health records, large healthcare datasets can now be analysed using predictive techniques to detect patterns and risk factors more accurately.

Therefore, the main problem addressed in this study is to analyse hospital data of diabetic patients and develop predictive insights that help identify factors associated with hospital readmission. By understanding these risk factors, healthcare providers can take preventive measures, improve patient monitoring, and reduce avoidable readmissions.

V. DATASET DESCRIPTION:

This study uses the Diabetes 130-US Hospitals dataset, which contains hospital admission records of diabetic patients collected over a period of ten years (1999–2008). The dataset includes more than 100,000 patient encounters from multiple hospitals and provides detailed clinical and demographic information.

A. Key characteristics of the dataset

Large-scale real hospital data, Over 100,000 patient records, Information collected from 130 hospitals, Data covering a 10year period, Contains missing and unclean values (raw clinical data),Important features included, Patient demographics (age, gender, race), Admission and discharge information, Medical diagnoses, Laboratory test results, Medications prescribed, Number of hospital visits, Length of hospital stay, Presence of other medical conditions, Readmission status (within 30 days, after 30 days, or no readmission), Target variable, The main outcome variable in this dataset is hospital readmission, which indicates whether a patient was readmitted within a specific time after discharge. This variable is used for predictive analysis.

B. Dataset Link

https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008

VI. REFERENCES:

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